

PAPER_586 — Big Bang Expansion Dynamics in UQFF Framework

Unified Quantum Field Framework (UQFF)

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CP4 Class: [#173 UQFFBigBangExpansionDynamicsCalculator](#) **Session:** 157 **Cross-refs:** PAPER_583 (6-Form), PAPER_587 (Inflation), PAPER_589 (Dark Energy) **Source:** grok_share_4cef778c78b8.txt

§1 Abstract

UQFF derives Big Bang initiation, 26-shell formation, and mass-buildup dynamics from the grinding mechanism and void buoyancy. The master BB equation produces an explicit scale factor $a(t)$ exhibiting accelerated expansion, dark energy, and the full primordial mass hierarchy — all from first principles without free cosmological parameters.

§2 Big Bang Initiation

Initial void state: $\rho \rightarrow 0$, $U_b \rightarrow +\infty$ (pure repulsion), SCm_{inj} injected at $t = 0$.

BB initiation:

$$BB_{\text{init}} = SCm_{inj} \cdot UA_{\text{contact}} \cdot \exp(\text{Grind}_{\text{opp}})$$

where $\text{Grind}_{\text{opp}} = \omega_{CW} \cdot SCm - \omega_{CCW} \cdot UA' \cdot e^{-H/v_{\text{init}}}$ encodes the CW/CCW angular momentum imbalance driving expansion.

§3 Complete Big Bang Equation (26-Shell Sum)

$$BB = S_{inj} \cdot U_{contact} \cdot \sum_{k=1}^{26} S_k^{26} \cdot \exp(\text{Grind}_{opp})$$

For uniform shell energies ($S_k = S$ for all k):

$$BB = 26 \cdot S_{inj} \cdot S^{26} \cdot U_{contact} \cdot \exp(\text{Grind}_{opp})$$

The S^{26} factor (26th power of primordial shell energy) quantizes cosmological expansion into exactly 26 discrete contributions.

§4 Adjusted Time

$$t_{adj} = t_{neg} + \frac{t_{obs}}{\Delta_{dil} + 1}$$

where $t_{neg} < 0$ is the negative-time pre-mass reservoir (see PAPER_597), $\Delta_{dil} = 0.1$ is the dilation factor, and t_{obs} is the observed cosmic time-scale.

§5 Pressure Order and Mass Buildup

$$P = \frac{(v_{init} - v_{current})(\Delta_{dil} t_{neg} + 1)}{\exp(-\mathcal{H}/v_{init}) \cdot \text{Partition}}$$

As $v_{current} \rightarrow v_{init}$ (mass builds up, velocity stabilizes), $P \rightarrow 0$ — the universe settles into its current expansion state.

§6 Scale Factor

Expansion catch-up velocity:

$$v_{exp} = (v_{init} - v_{current}) \cdot \frac{\exp(\text{Grind})}{t_{adj}}$$

Scale factor (power-law derived from catch-up):

$$a(t) = t^{-(v_{current} - v_{init}) \exp(\text{Grind})}$$

For $v_{init} > v_{current}$: the exponent is positive $\Rightarrow a(t)$ increases with t (expansion). For early times ($t \sim 10^{-32}$ s) with rapid Grind, expansion is super-exponential.

§7 Primordial Hydrogen Formation

$$E_{\text{ProtoH}} = \sum_{k=1}^{26} E_{\text{Grind}}(k) + \int_{t_{\text{adj}}}^{t_{\text{H}}}$$

- $\cdot E_{\text{Higgs}} \cdot \sum_k E_{\text{ShellEnergies}}(k)$

The 26-shell integral of Grind produces the binding energy needed for neutral hydrogen formation (recombination epoch at $t \sim 380,000$ yr).

§8 Comparison to Λ CDM

Observable	Λ CDM	UQFF BB
Scale factor $a(t)$	$t^{2/3}$ (matter)	$t^{-(v_c-v_i)\exp(G)}$
Dark energy	Free parameter Λ	Derived from U_b
Singularity at $t=0$	Yes ($a \rightarrow 0$)	No ($r_{\text{min}} > 0$)
Shell structure	None	26 discrete shells
CMB temperature	Fit	From $\sum_{k=1}^{26} S_k$

§9 Conclusions

UQFF derives Big Bang dynamics from first principles: grinding imbalance drives expansion, 26-shell quantization produces the mass hierarchy, and the S_{26} bound prevents the initial singularity. The scale factor reproduces Λ CDM behavior as a limiting case.

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